

Forecasting Beijing PM2.5 Six Hours Ahead

Gradient boosting with rolling chronological backtests

Project 19 · Semester 5 Applied Machine Learning

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Why forecast PM2.5?

Serious health risk

Fine particles reach deep into the lungs and are linked to heart and lung disease.

Act before the peak

A 6-hour warning lets people limit outdoor activity and lets cities respond.

Test ML fairly

Does a model beat simple rules when tested only on the future?

Our setup

Target	**PM2.5 (µg/m³)**
Horizon	**6 hours ahead**
Main model	**XGBoost**
Baselines	**Persistence, seasonal naive**
Evaluation	**Rolling backtest, 4 folds**
Metrics	**MAE, RMSE, R^2**
AQI standard	**US EPA (2024)**

Literature review

Study	Approach	Key finding for us
Liang et al. (2015)	Statistical study, Beijing 2010–2015	Weather and winter heating drive PM2.5; source of our dataset
Rybarczyk & Zalakeviciute (2018)	Review of 46 air quality studies	ML handles non-linear pollution better than deterministic models
Pan (2018)	XGBoost, hourly PM2.5, Tianjin	XGBoost beat RF, linear regression, decision tree and SVR
Zhai & Chen (2018)	Stacked ensemble with XGBoost, Beijing	EDA-driven features with tree models work well
Li et al. (2017)	LSTM with spatiotemporal data	Sequence models learn temporal patterns
Huang & Kuo (2018)	CNN-LSTM on the same Beijing data	Deep learning reference on our dataset
Tashman (2000); Bergmeir & Benítez (2012)	Forecast evaluation methods	Test with rolling-origin, order-preserving splits

Full references in IEEE format are in the project report.

Research gap and our approach

What we noticed

- Hand-picked lag windows may miss longer patterns and pollution–weather interactions
- A model is only useful if it beats simple rules such as persistence
- Random splits leak the future into training

Our approach

- XGBoost captures non-linear interactions
- Compared against persistence and seasonal naive baselines
- Rolling chronological backtest with a leakage guard
- Stated AQI standard (US EPA 2024); LSTM as the next step

The dataset

43,824
hourly records

5 years
2010 – 2014

8
variables

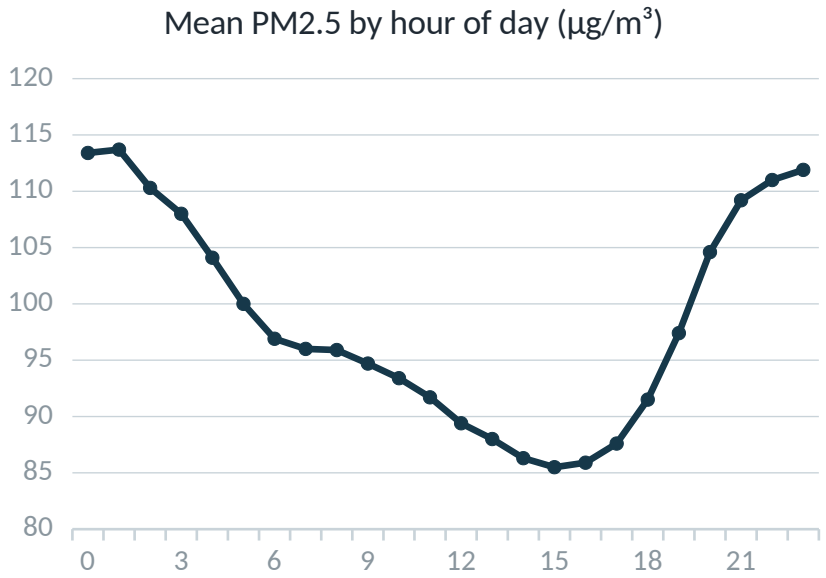
4.7%
PM2.5 hours missing

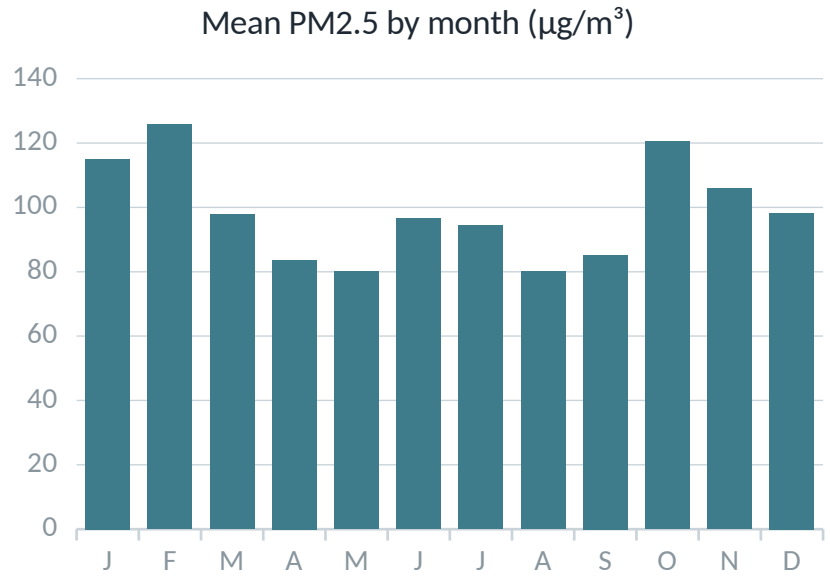
Variables

- **PM2.5 (target)**
- Dew point
- Temperature
- Pressure
- Wind direction
- Wind speed
- Snow hours
- Rain hours

Source: UCI Machine Learning Repository (dataset 381), US Embassy monitor in Beijing

Pollution peaks at night and in winter

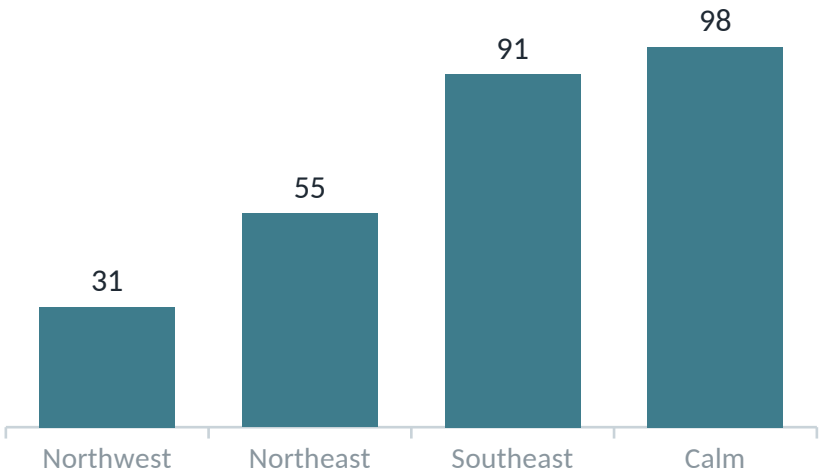




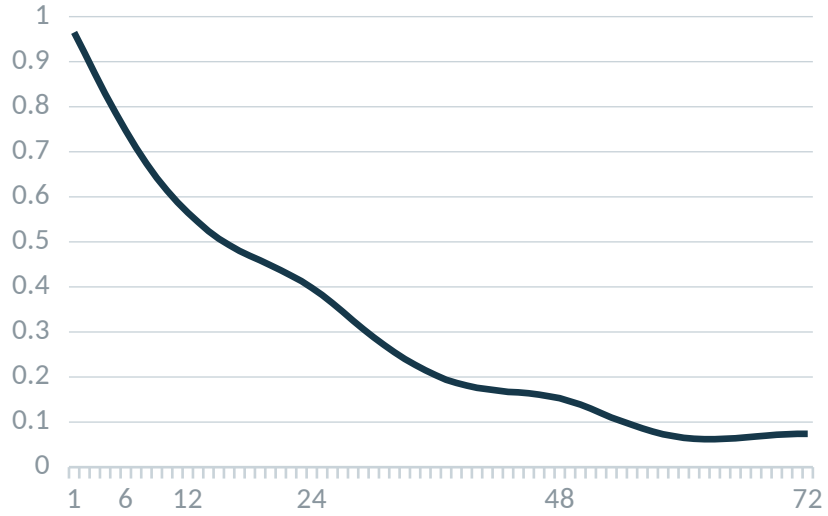
So what: we add hour, month and their sin/cos encodings as calendar features.

PM2.5 is persistent and wind-driven

Median PM2.5 by wind direction (µg/m³)

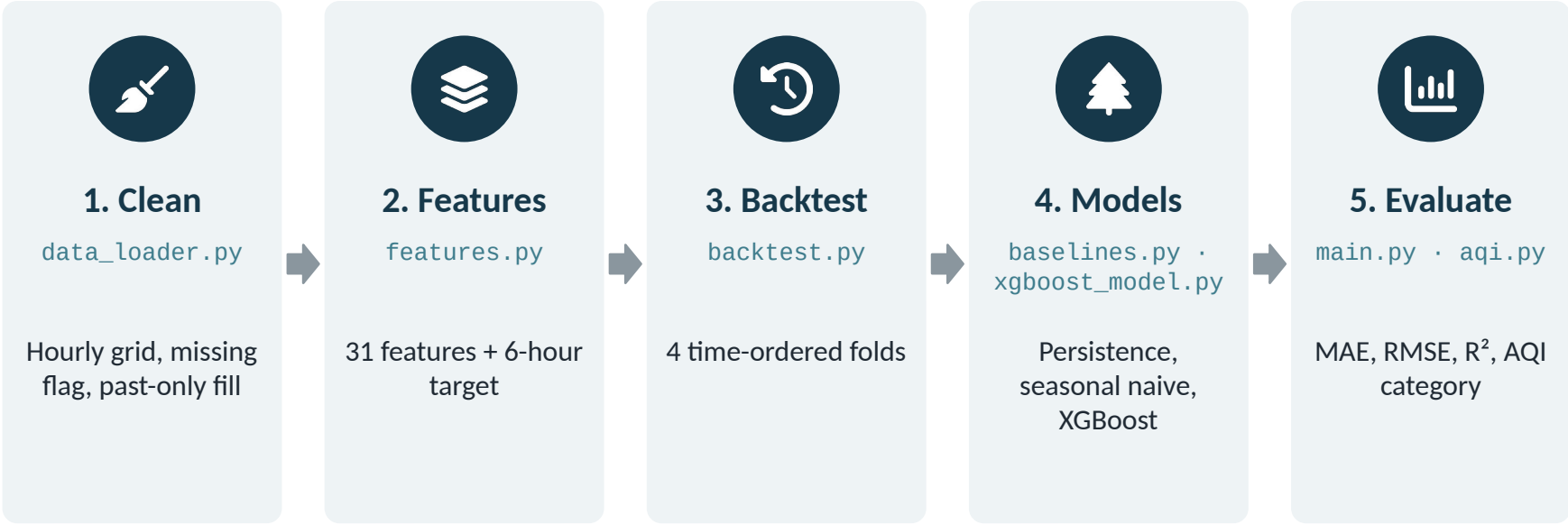


Correlation with PM2.5 k hours earlier



So what: northwest wind clears the air, and PM2.5 still correlates 0.75 with 6 hours earlier, so we add wind columns and lag features.

Our pipeline



Each step is one small Python file owned by one team member. Notebooks, 14 figures and 10 automated tests are in the public GitHub repository.

31 features in 7 groups

1

Current PM2.5

PM2.5 right now

7

Lags

1, 2, 3, 6, 12, 24 h ago + same hour yesterday

2

Rolling means

Averages of the last 6 h and 24 h

8

Calendar

Hour, weekday, month, weekend, sin/cos

8

Weather

Temperature, dew point, pressure, wind speed, rain, snow + 2 derived

4

Wind direction

NE, NW, SE, calm (one-hot)

1

Missing flag

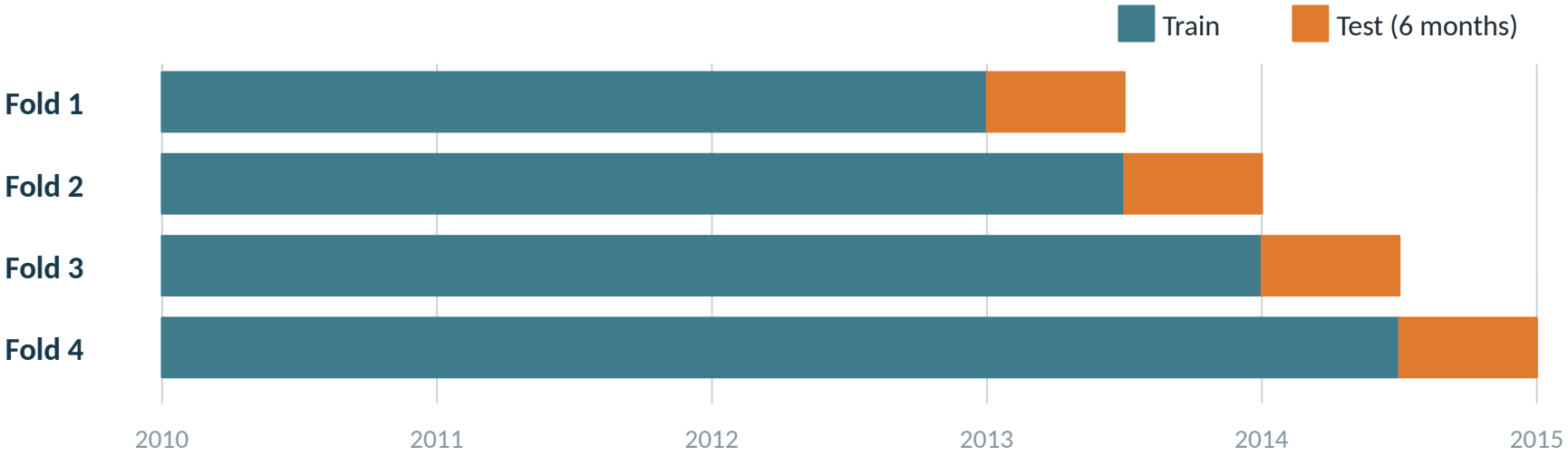
Was PM2.5 measured this hour?

6 h

Target

PM2.5 6 hours later, measured hours only

A fair, time-ordered backtest



Future only

Train on all earlier data, test on the next 6 months. Never shuffle.

No leakage

The last 6 training hours are never used as targets.

Same test for all

Every model is scored on the same measured hours.

Three models compared

BASELINE

Persistence

PM2.5 in 6 hours = PM2.5 now

Strong because pollution changes slowly

BASELINE

Seasonal naive

PM2.5 in 6 hours = PM2.5 at the same hour yesterday

Uses the daily cycle

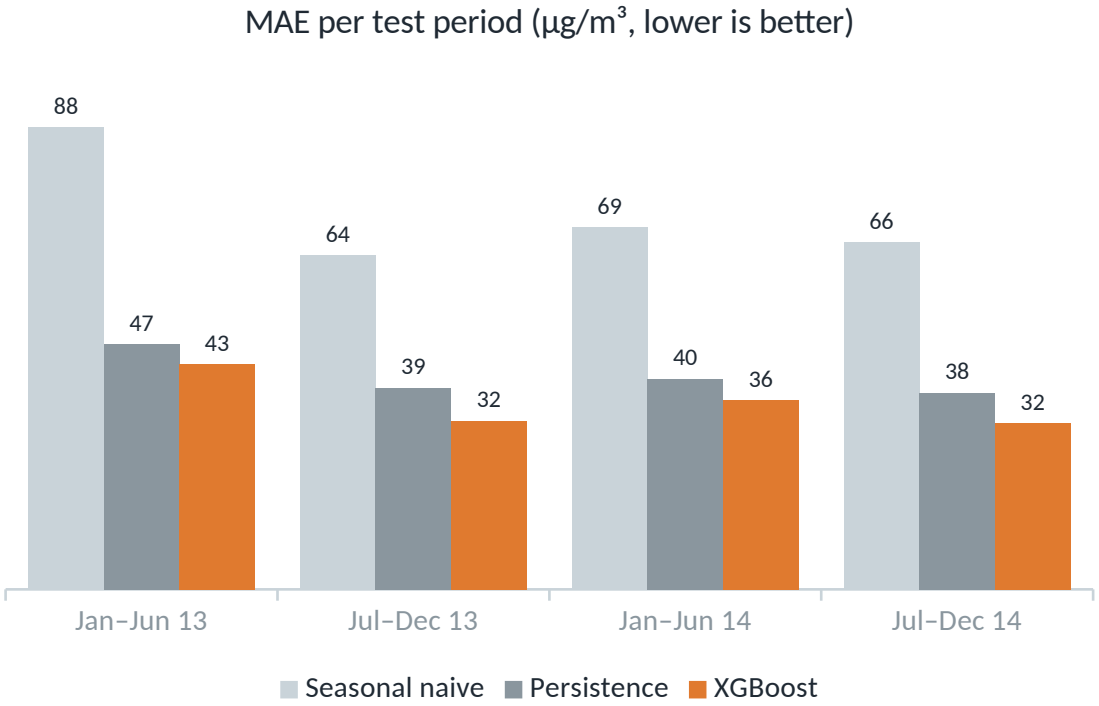
OUR MODEL

XGBoost

300 gradient-boosted trees, depth 6, learning rate 0.05, 80% row and column sampling

Learns non-linear weather × time effects

XGBoost wins in every test period



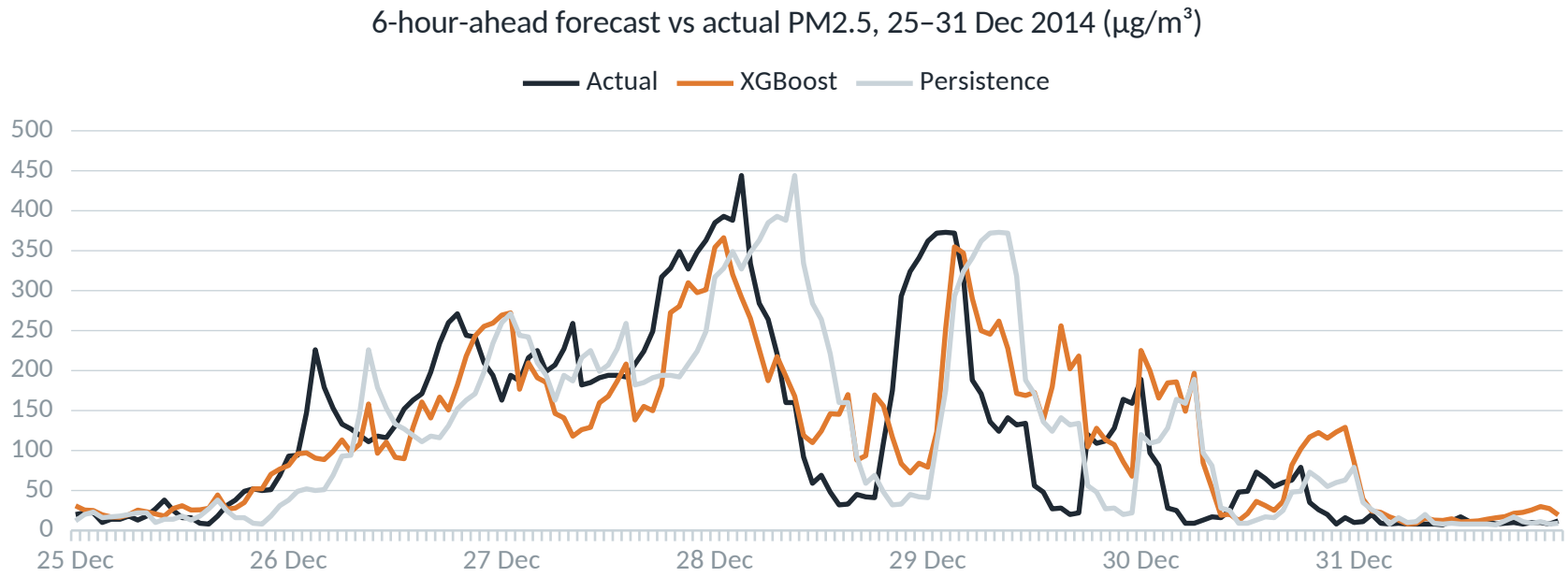
12%

lower error than persistence (average MAE)

Average	MAE	R^2
Seasonal naive	71.8	-0.22
Persistence	40.7	0.52
XGBoost	**35.8**	**0.65**

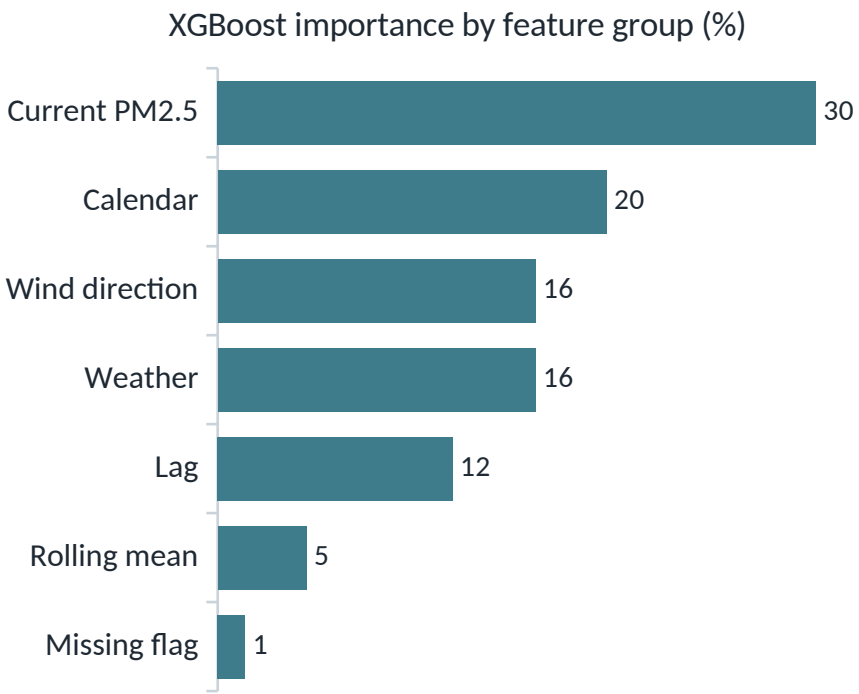
AQI category correct: XGBoost 53.9% vs persistence 49.8%

Forecasts track real pollution



Read it: persistence always arrives 6 hours late. XGBoost often turns earlier, but sudden spikes remain hard to predict.

What drives the model



Current PM2.5 · 30%

Pollution now is the best single clue.

Calendar · 20%

Night and winter peaks matter.

Wind and weather · 32%

Northwest wind clears the air.

Lags · only 12%

Room for a sequence model (LSTM).

Limitations and next steps

Limitations

- Sudden spikes have no early warning in 6 hours of history
- Lags are hand-picked and stop at 24 hours
- One station, 2010–2014 only
- 24-hour AQI breakpoints applied to hourly values

Next steps

- LSTM on the last 48 raw hours
- Add weather forecasts and nearby stations
- Tune on a time-ordered validation split
- Forecast 24-hour averages for a standard AQI

Key takeaways

12%

lower error than persistence (MAE 35.8 vs 40.7 µg/m^3)

4 / 4

test periods won by XGBoost under a fair, future-only backtest

10 / 10

automated tests pass; 6 notebooks and 14 figures regenerate every number



github.com/MananGilhotra/air-quality-forecasting

Thank you · Questions?